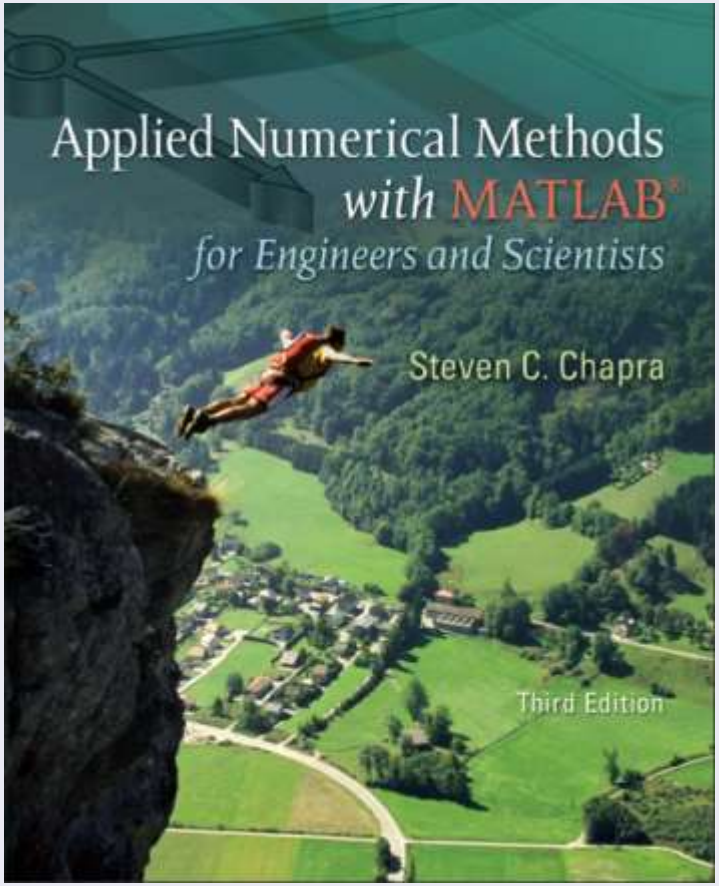


Part 3
Chapter 13

Eigenvalues

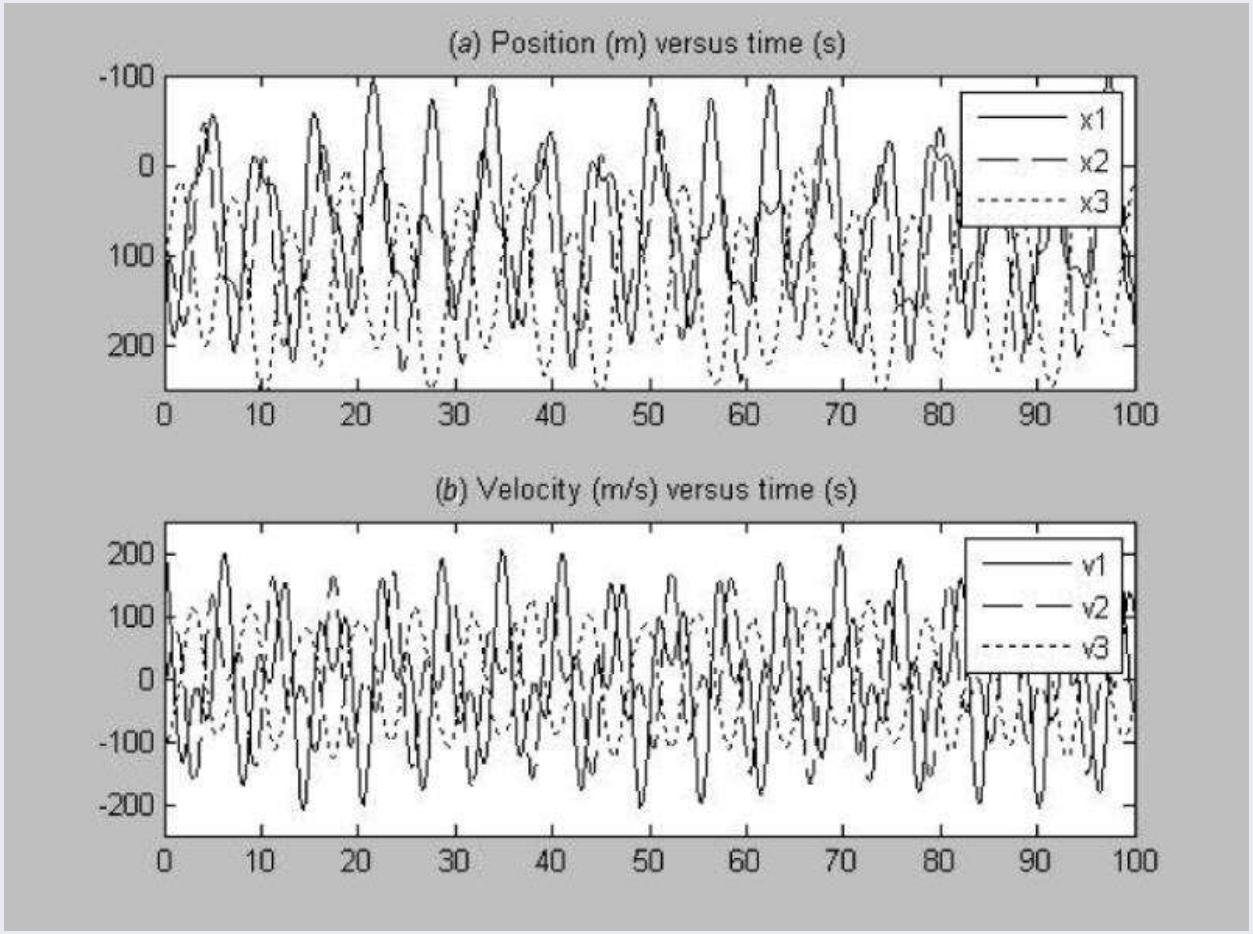
PowerPoints organized by Prof. Steve Chapra, Tufts University

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Chapter Objectives

- Understanding the mathematical definition of eigenvalues and eigenvectors.
- Understanding the physical interpretation of eigenvalues and eigenvectors within the context of engineering systems that vibrate or oscillate.
- Knowing how to implement the polynomial method.
- Knowing how to implement the power method to evaluate the largest and smallest eigenvalues and their respective eigenvectors.
- Knowing how to use and interpret MATLAB's **`eig`** function.

Dynamics of Three Coupled Bungee Jumpers in Time



Is there an underlying pattern???

Mathematics

Up until now, heterogeneous systems:

$$[A]\,\{x\} = \{b\}$$

What about homogeneous systems:

$$[A]\,\{x\} = 0$$

Trivial solution:

$$\{x\} = 0$$

Is there another way of formulating the system so that the solution would be meaningful???

Mathematics

What about a homogeneous system like:

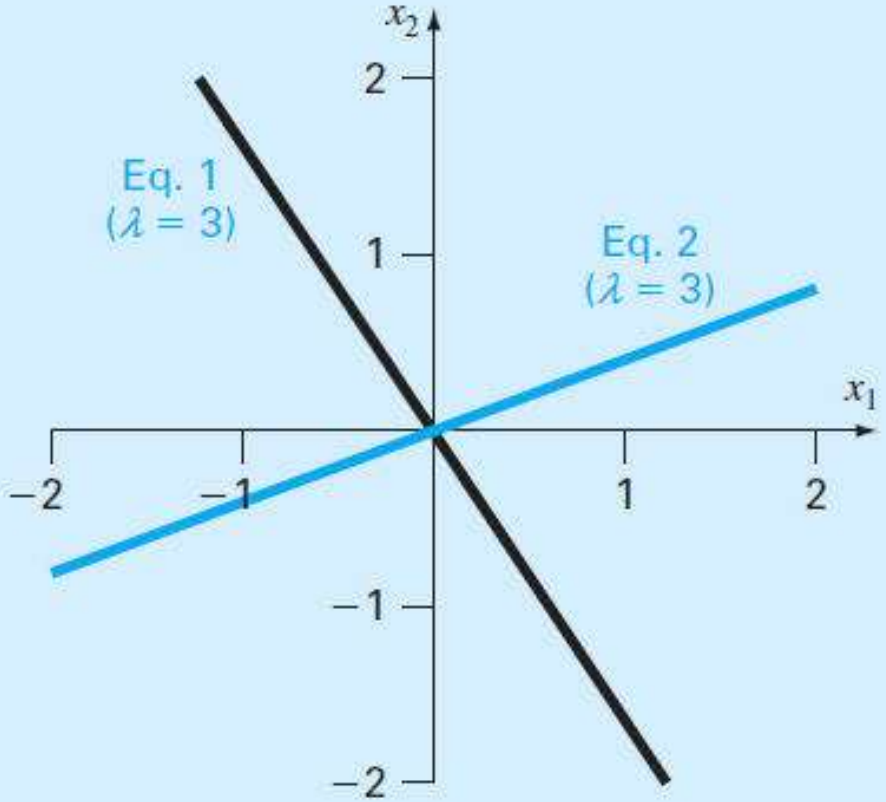
$$\begin{aligned}(a_{11}-\lambda)\,x_1 + a_{12}\,x_2 + a_{13}\,x_3 &= 0\\ a_{21}\,x_1 + (a_{22}-\lambda)\,x_2 + a_{23}\,x_3 &= 0\\ a_{31}\,x_1 + a_{32}\,x_2 + (a_{33}-\lambda)\,x_3 &= 0\end{aligned}$$

or in matrix form

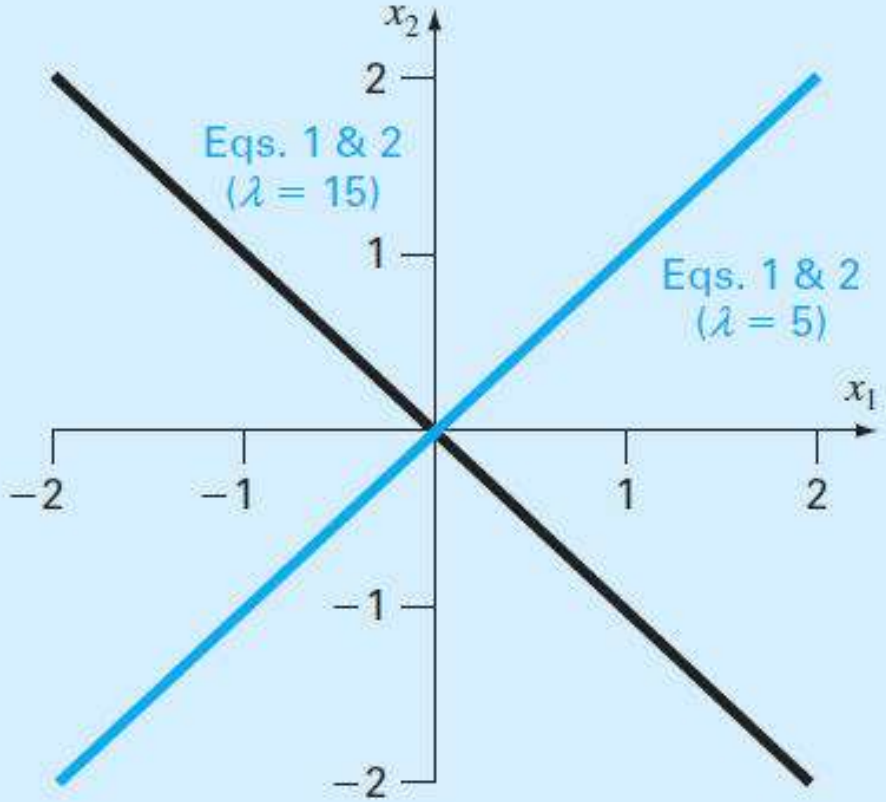
$$\Big[[A]-\lambda\,[I]\Big]\{x\}=0$$

For this case, there could be a value of λ that makes the equations equal zero. This is called an ***eigenvalue***.

Graphical Depiction of Eigenvalues

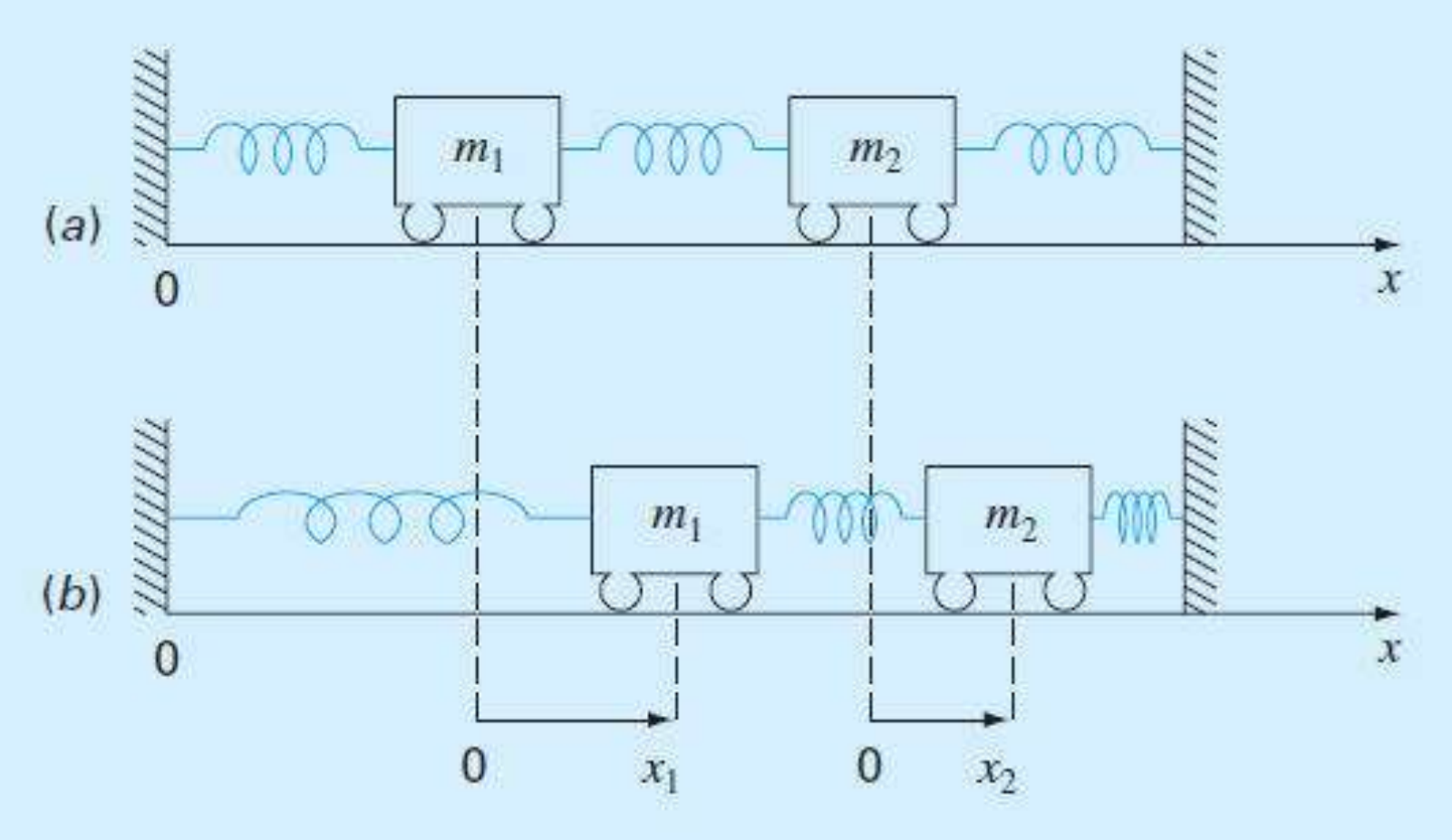


(*a*) Incorrect eigenvalue



(*b*) Correct eigenvalues

Physical Background: Oscillations or ***Vibrations*** of Mass-Spring Systems



Model With Force Balances (AKA: $F = ma$)

$$m_1 \frac{d^2 x_1}{dt^2} = -k\,x_1 + k(x_2 - x_1)$$

$$m_2 \frac{d^2 x_2}{dt^2} = -k(x_2 - x_1) - kx_2$$

Collect terms:

$$m_1 \frac{d^2 x_1}{dt^2} - k\,(-2x_1 + x_2) = 0$$

$$m_2 \frac{d^2 x_2}{dt^2} - k\,(x_1 - 2x_2) = 0$$

Assume a Sinusoidal Solution

$$x_i = X_i \sin(\omega t) \qquad \text{where} \qquad \omega = \frac{2\pi}{T_p}$$

Differentiate twice:

$$x_i'' = -X_i \omega^2 \sin(\omega t)$$

Substitute back into system and collect terms

$$\left(\frac{2k}{m_1} - \omega^2 \right) X_1 - \frac{k}{m_1} X_2 = 0$$

$$\frac{k}{m_2} X_1 + \left(\frac{2k}{m_2} - \omega^2 \right) X_2 = 0$$

Given: $m_1 = m_2 = 40$ kg; $k = 200$ N/m

$$(10 - \omega^2) X_1 - 5 X_2 = 0$$

$$-5 X_1 + (10 - \omega^2) X_2 = 0$$

This is now a homogeneous system where the eigenvalue represents the square of the fundamental frequency.

Solution: The Polynomial Method

$$\begin{bmatrix} 10-\omega^2 & -5 \\ -5 & 10-\omega^2 \end{bmatrix} \begin{Bmatrix} X_1 \\ X_2 \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix}$$

Evaluate the determinant to yield a polynomial

$$\begin{vmatrix} 10-\omega^2 & -5 \\ -5 & 10-\omega^2 \end{vmatrix} = (\omega^2)^2 - 20\omega^2 + 75$$

The two roots of this "characteristic polynomial" are the system's eigenvalues:

$$\omega^2 = \begin{matrix} 15 \\ 5 \end{matrix} \quad \text{or} \quad \omega = \begin{matrix} 3.873 \text{ Hz} \\ 2.36 \text{ Hz} \end{matrix}$$

INTERPRETATION

$\omega^2 = 5\ /s^2$

$\omega = 2.236\ /s$

$T_p = 2\pi/2.236 = 2.81\ s$

$\omega^2 = 15\ /s^2$

$\omega = 3.873\ /s$

$T_p = 2\pi/3.373 = 1.62\ s$

$$(10 - \omega^2)\, X_1 - 5\, X_2 = 0$$
$$-5\, X_1 + (10 - \omega^2)\, X_2 = 0$$

$$(10 - 5)\, X_1 - 5\, X_2 = 0$$
$$-5\, X_1 + (10 - 5)\, X_2 = 0$$

$$5\, X_1 - 5\, X_2 = 0$$
$$-5\, X_1 + 5\, X_2 = 0$$

$$X_1 = X_2$$

$$V = \begin{bmatrix} -0.7071 \\ -0.7071 \end{bmatrix}$$

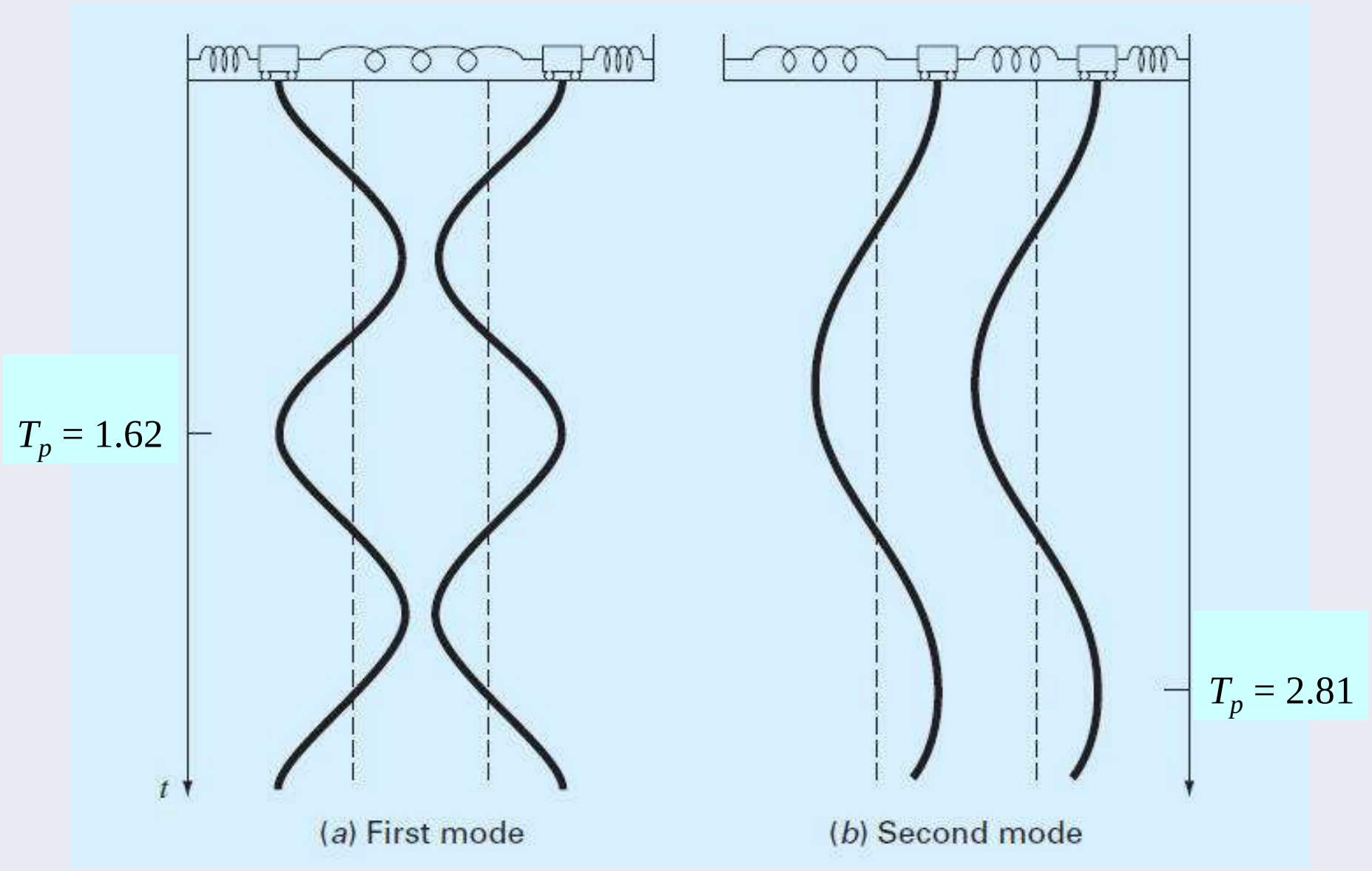
$$(10 - 15)\, X_1 - 5\, X_2 = 0$$
$$-5\, X_1 + (10 - 15)\, X_2 = 0$$

$$-5\, X_1 - 5\, X_2 = 0$$
$$-5\, X_1 - 5\, X_2 = 0$$

$$X_1 = -X_2$$

$$V = \begin{bmatrix} -0.7071 \\ 0.7071 \end{bmatrix}$$

Principle Modes of Vibration



The Power Method

Iterative method to compute the largest eigenvalue ***and*** its associated eigenvector.

$$[[A] - \lambda[I]]]\{x\} = 0$$

$$[A]\{x\} = \lambda\{x\}$$

Simple Algorithm:

```
function [eval, evect] = powereig(A,es,maxit)
n=length(A);
evect=ones(n,1);eval=1;iter=0;ea=100; %initialize
while(1)
  evalold=eval;              %save old eigenvalue value
  evect=A*evect;             %determine eigenvector as [A]*{x)
  eval=max(abs(evect));      %determine new eigenvalue
  evect=evect./eval;         %normalize eigenvector to eigenvalue
  iter=iter+1;
  if eval~=0, ea = abs((eval-evalold)/eval)*100; end
  if ea<=es | iter >= maxit,break,end
end
```

Example: The Power Method

First iteration:

$$\begin{bmatrix} 40 & -20 & 0 \\ -20 & 40 & -20 \\ 0 & -20 & 40 \end{bmatrix} \begin{Bmatrix} 1 \\ 1 \\ 1 \end{Bmatrix} = \begin{Bmatrix} 20 \\ 0 \\ 20 \end{Bmatrix} = 20 \begin{Bmatrix} 1 \\ 0 \\ 1 \end{Bmatrix}$$

Second iteration:

$$\begin{bmatrix} 40 & -20 & 0 \\ -20 & 40 & -20 \\ 0 & -20 & 40 \end{bmatrix} \begin{Bmatrix} 1 \\ 0 \\ 1 \end{Bmatrix} = \begin{Bmatrix} 40 \\ -20 \\ 40 \end{Bmatrix} = 40 \begin{Bmatrix} 1 \\ -1 \\ 1 \end{Bmatrix}$$

$$|\varepsilon_a| = \left| \frac{40 - 20}{40} \right| \times 100\% = 50\%$$

Example: The Power Method

Third iteration:

$$\begin{bmatrix} 40 & -20 & 0 \\ -20 & 40 & -20 \\ 0 & -20 & 40 \end{bmatrix} \begin{Bmatrix} 1 \\ -1 \\ 1 \end{Bmatrix} = \begin{Bmatrix} 60 \\ -80 \\ 60 \end{Bmatrix} = -80 \begin{Bmatrix} -0.75 \\ 1 \\ -0.75 \end{Bmatrix}$$

$$|\varepsilon_a| = \left| \frac{-80 - 40}{-80} \right| \times 100\% = 150\%$$

Fourth iteration:

$$\begin{bmatrix} 40 & -20 & 0 \\ -20 & 40 & -20 \\ 0 & -20 & 40 \end{bmatrix} \begin{Bmatrix} -0.75 \\ 1 \\ -0.75 \end{Bmatrix} = \begin{Bmatrix} -50 \\ 75 \\ -50 \end{Bmatrix} = 70 \begin{Bmatrix} -0.71429 \\ 1 \\ -0.71429 \end{Bmatrix}$$

$$|\varepsilon_a| = \left| \frac{70 - (-80)}{70} \right| \times 100\% = 214\%$$

Example: The Power Method

Fifth iteration:

$$\begin{bmatrix} 40 & -20 & 0 \\ -20 & 40 & -20 \\ 0 & -20 & 40 \end{bmatrix} \begin{Bmatrix} -0.71429 \\ 1 \\ -0.71429 \end{Bmatrix} = \begin{Bmatrix} -48.51714 \\ 68.51714 \\ -48.51714 \end{Bmatrix} = 68.51714 \begin{Bmatrix} -0.71429 \\ 1 \\ -0.71429 \end{Bmatrix}$$

$$|\varepsilon_a| = \left| \frac{68.51714 - 70}{70} \right| \times 100\% = 2.08\%$$

The process can be continued to determine the largest eigenvalue (= 68.284) with the associated eigenvector [–0.7071 1 –0.7071]

Note that the smallest eigenvalue and its associated eigenvector can be determined by applying the power method to the inverse of A

Determining Eigenvalues & Eigenvectors with MATLAB

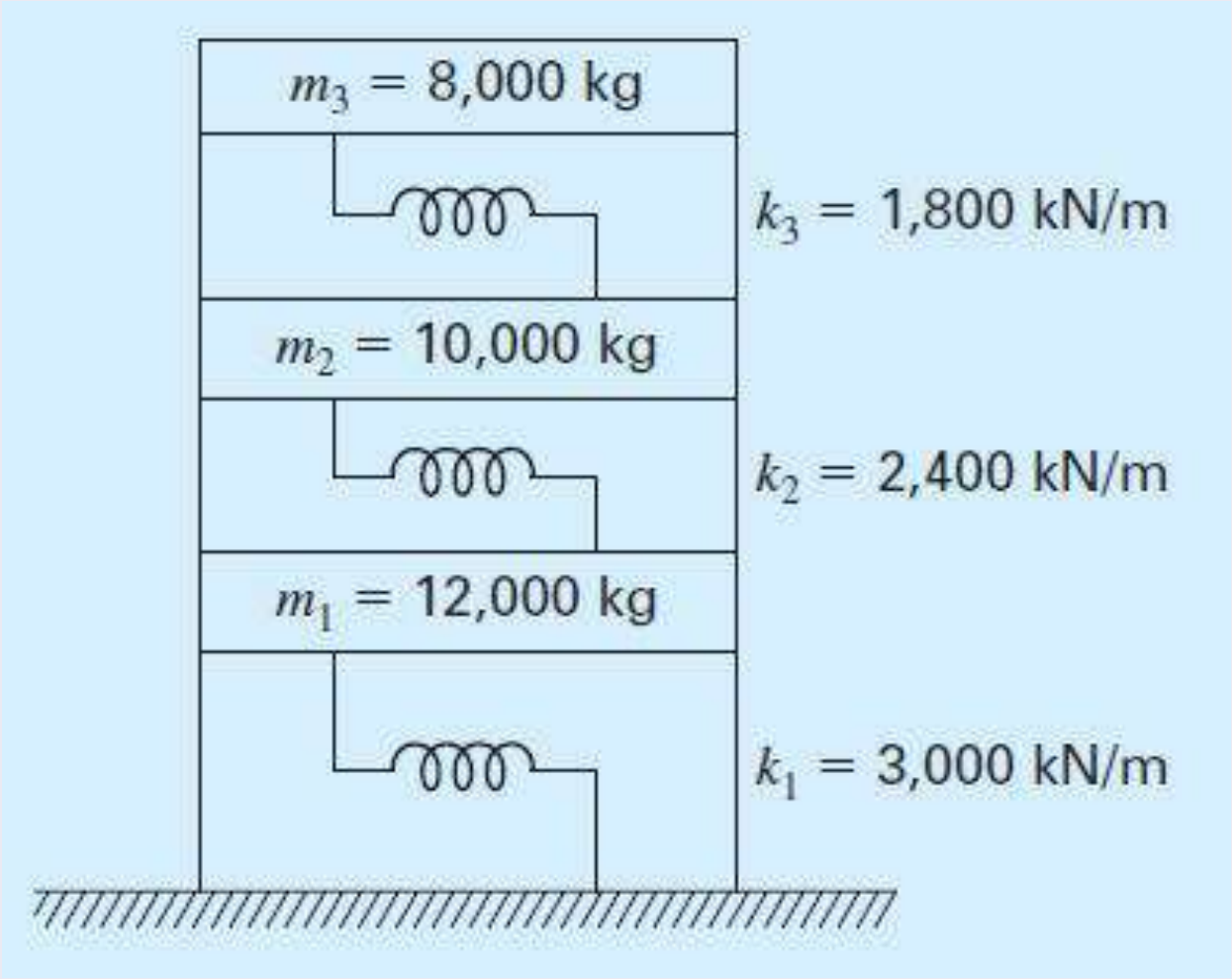
```
>> A = [10 -5;-5 10]

A =
    10    -5
    -5    10

>> [v,lambda] = eig(A)

v =
   -0.7071   -0.7071
   -0.7071    0.7071
lambda =
     5     0
     0    15
```

Dynamics of Three Story Building



Principle Modes of Vibration

